8.3 Trigonometric Integrals

Trigonometric integrals involve algebraic combinations of the six basic trigonometric functions. The general idea is to use identities to transform the integrals we need to evaluate into integrals that are easier to work with.

To evaluate trigonometric integrals, we may apply some techniques of integration such as basic formulas, basic substitutions, integration by parts, using identities, multiplying an integrand by a form of 1 and eliminate square roots.

Example 1: (Use Identities)

$$\int(\sec x+\tan x)^2\,dx \;=\int(\sec^2 x+2\sec x\tan x+\tan^2 x)dx$$

$$=\int(\sec^2 x+2\sec x\tan x+\sec^2 x-1)dx \;=2\tan x+2\sec x-x+C$$

Example 2: (Eliminate Square roots)

1. $$\int_0^{\pi}\sqrt{1-\cos 4x}\,dx \;=\int_0^{\pi}\sqrt{2\sin^2 2x}\,dx \;=\sqrt{2}\int_0^{\pi}|\sin 2x|dx$$

$$=\sqrt{2}\left(\int_0^{\pi/2}\sin 2x dx-\int_{\pi/2}^{\pi}\sin 2x dx\right)=\sqrt{2}\left(-\frac{\cos 2x}{2}\Big]_0^{\pi/2}+\frac{\cos 2x}{2}\Big]_{\pi/2}^{\pi}\right)=2\sqrt{2}$$

2. $$\int_0^{\frac{\pi}{4}}\sqrt{1+\cos 4x}\,dx=\int_0^{\frac{\pi}{4}}\sqrt{2\cos^2 2x}\,dx$$

$$=\sqrt{2}\int_0^{\frac{\pi}{4}}|\cos 2x|\,dx=\sqrt{2}\int_0^{\frac{\pi}{4}}\cos 2x\,dx \qquad (\because\ \cos 2x\ \geq 0 \text{ on } [0,\tfrac{\pi}{4}])$$

$$=\frac{\sqrt{2}}{2}[\sin 2x]_0^{\frac{\pi}{4}}=\frac{\sqrt{2}}{2}\left(\sin\frac{\pi}{2}-\sin 0\right)=\frac{\sqrt{2}}{2}$$

1. Products of Powers of Sines and Cosines

Integrals of the form $\int\sin^m x\cos^n x dx$, where *m* and *n* are nonnegative integers.

Case 1: *m* is odd or *n* is odd

(1.1) When *m* is odd and *n* is even

Write $\sin^m x=\sin^{m-1}\sin x$ and use the identity $\sin^2 x=1-\cos^2 x$.
Then solve the integral by substituting $u = \cos x$.

(1.2) When *n* is odd and *m* is even

Write $\cos^n x=\cos^{n-1}\cos x$ and use the identity $\cos^2 x=1-\sin^2 x$.
Then solve the integral by substituting $u = \sin x$.

(1.3) When *m* and *n* are odd

Both methods in (1.1) and (1.2) can be applied.

Case 2: Both *m* and *n* are even

Use the identities $\sin^2 x=\frac{1}{2}(1-\cos 2x)$ and $\cos^2 x=\frac{1}{2}(1+\cos 2x)$.
To reduce the integrand to one in lower powers of cos2*x*.

Example 3: (*m* is odd or *n* is odd)

Evaluate (1) $\int \sin^3 x \cos^2 x\, dx$ (2) $\int_0^{\frac{\pi}{2}} 35\sin^4 x \cos^3 x\, dx$

(3) $\int \cos^5 x\, dx$ (4) $\int \sin^7 x \cos^5 x\, dx$

Solution:

(1) $\int \sin^3 x \cos^2 x\, dx = \int \sin^2 x \cos^2 x \sin x\, dx$ power of sin$x$ is odd

$= \int (1-\cos^2 x)\cos^2 x \sin x\, dx$ use $\sin^2 x = 1-\cos^2 x$

$= \int (1-u^2)\,u^2(-du)$ (Let $u = \cos x$. Then $du = -\sin x dx$.)

$= \int (u^4 - u^2)du = \frac{u^5}{5} - \frac{u^3}{3} + C = \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$

(2) $\int_0^{\pi/2} 35\sin^4 x\cos^3 x dx = 35\int_0^{\pi/2} \sin^4 x\cos^2 x\cos x dx$ power of cos$x$ is odd

$= 35\int_0^{\pi/2} (\sin^4 x)(1-\sin^2 x)\cos x dx$ (Let $u = \sin x$. Then $du = \cos x dx$.

When $x = 0$, $u = 0$; $x = \pi/2$, $u = 1$.)

$= 35\int_0^1 u^4(1-u^2)du = 35\int_0^1 (u^4 - u^6)du = \left[7u^5 - 5u^7\right]_0^1 = 2$

(3) $\int \cos^5 x\, dx = \int \cos^4 x \cos x\, dx$

$= \int (1-\sin^2 x)^2 \cos x\, dx$

$= \int (1-u^2)^2 du$ (Let $u = \sin x$. Then $du = \cos x dx$.)

$= \int (1 - 2u^2 + u^4)\, du$

$= u - \frac{2}{3}u^3 + \frac{1}{5}u^5 + C = \sin x - \frac{2}{3}\sin^3 x + \frac{1}{5}\sin^5 x + C$

(4) $\int \sin^7 x\cos^5 x dx = \int \sin^7 x\cos^4 x\cos x dx = \int (\sin^7 x)(\cos^2 x)^2 \cos x dx$

$= \int (\sin^7 x)(1-\sin^2 x)^2 \cos x dx$ (Let $u = \sin x$. Then $du = \cos x dx$.)

$= \int u^7(1-2u^2+u^4)du = \int (u^7 - 2u^9 + u^{11})du$

$= \frac{u^8}{8} - \frac{u^{10}}{5} + \frac{u^{12}}{12} + C = \frac{1}{120}\left(15\sin^8 x - 24\sin^{10} x + 10\sin^{12} x\right) + C$

Alternative method:

$\int \sin^7 x\cos^5 x dx = \int \sin^6 x \sin x \cos^5 x dx$

$= \int (\sin^2 x)^3 \cos^5 x \sin x dx = \int (1-\cos^2 x)^3 \cos^5 x \sin x dx$ (Let $u = \cos x$. Then $du = -\sin x dx$.)

$= -\int u^5(1-u^2)^3 du = -\int u^5(1-3u^2+3u^4-u^6)du = -\int (u^5 - 3u^7 + 3u^9 - u^{11})du$

$= -\frac{u^6}{6} + \frac{3u^8}{8} - \frac{3u^{10}}{10} + \frac{u^{12}}{12} + C = -\frac{1}{120}\left(20\cos^6 x - 45\cos^8 x + 36\cos^{10} x - 10\cos^{12} x\right) + C$

Example 4: (Both *m* and *n* are even)

Evaluate (1) $\int \sin^2 x \cos^4 x\, dx$ (2) $\int \cos^6 x\, dx$

Solution:

(1) $\int \sin^2 x \cos^4 x\, dx = \int \left(\frac{1-\cos 2x}{2}\right)\left(\frac{1+\cos 2x}{2}\right)^2 dx$

$= \frac{1}{8}\int (1-\cos 2x)(1+2\cos 2x+\cos^2 2x)\, dx$

$= \frac{1}{8}\int (1+\cos 2x-\cos^2 2x-\cos^3 2x)\, dx$

$= \frac{1}{8}\left[x+\frac{1}{2}\sin 2x-\int (\cos^2 2x+\cos^3 2x)\, dx\right]$

Use the identity $\cos^2\theta = (1+\cos 2\theta)/2$, with $\theta = 2x$.

$\int \cos^2 2x\, dx = \frac{1}{2}\int (1+\cos 4x)\, dx$

$= \frac{1}{2}\left(x+\frac{1}{4}\sin 4x\right)+C_1.$

$\int \cos^3 2x\, dx = \int (1-\sin^2 2x)\cos 2x\, dx$ (Let $u = \sin 2x$. Then $du = 2\cos 2x\, dx$.)

$= \frac{1}{2}\int (1-u^2)\, du = \frac{1}{2}\left(\sin 2x-\frac{1}{3}\sin^3 2x\right)+C_2.$

Therefore, $\int \sin^2 x \cos^4 x\, dx = \frac{1}{16}\left(x-\frac{1}{4}\sin 4x+\frac{1}{3}\sin^3 2x\right)+C.$

(2) $\int \cos^6 x\, dx = \int (\cos^2 x)^3\, dx = \int \left(\frac{1+\cos 2x}{2}\right)^3 dx$

$= \frac{1}{8}\int (1+3\cos 2x+3\cos^2 2x+\cos^3 2x)\, dx$

$= \frac{1}{8}\left[x+\frac{3\sin 2x}{2}+\frac{3}{2}\left(x+\frac{\sin 4x}{4}\right)+\frac{1}{2}\left(\sin 2x-\frac{\sin^3 2x}{3}\right)\right]+C$ (From part (1))

$= \frac{1}{8}\left[\frac{5}{2}x+2\sin 2x+\frac{3}{8}\sin 4x-\frac{\sin^3 2x}{6}\right]+C$

2. Integrals of tangent, cotangent, secant and cosecant functions

1. $\int \tan x dx = \ln\lvert\sec x\rvert + C$	2. $\int \sec x dx = \ln\lvert\sec x+\tan x\rvert + C$
3. $\int \cot x dx = \ln\lvert\sin x\rvert + C$	4. $\int \csc x dx = -\ln\lvert\csc x+\cot x\rvert + C$

Example 5: (Use Identities)

$\int \tan x dx = \int \frac{\sin x}{\cos x} dx = \int \frac{-du}{u}$ (Let $u = \cos x$. Then $du = -\sin x dx$.)

$= -\ln\lvert u\rvert + C \quad = \ln\lvert\cos x\rvert^{-1} + C = \ln\lvert\sec x\rvert + C$

Example 6: (Multiply by a form of 1)

$$\int \sec x\,dx = \int \sec x \left(\frac{\sec x + \tan x}{\sec x + \tan x}\right) dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x}\,dx$$ (Let $u = \sec x + \tan x$.

Then $du = (\sec x \tan x + \sec^2 x)\,dx$.

$= \int \frac{du}{u} = \ln|u| + C = \ln|\sec x + \tan x| + C$

3. Integrals of Powers of tan *x*, sec *x*, ccs *x* and cot *x*

Use the identities $1 + \tan^2 x = \sec^2 x$ and $1 + \cot^2 x = \csc^2 x$.
Integrate by substitution or by parts when necessary to reduce the powers to lower powers.

Example 7:

Evaluate (1) $\int \tan^4 x\,dx$ (2) $\int \sec^3 x\,dx$ (3) $\int \tan^4 x \sec^4 x\,dx$

Solution:

(1)
$$\begin{aligned}
\int \tan^4 x\,dx &= \int \tan^2 x \cdot \tan^2 x\,dx \\
&= \int \tan^2 x \cdot (\sec^2 x - 1)\,dx && \tan^2 x = \sec^2 x - 1 \\
&= \int \tan^2 x \sec^2 x\,dx - \int \tan^2 x\,dx \\
&= \int \tan^2 x \sec^2 x\,dx - \int (\sec^2 x - 1)\,dx && \tan^2 x = \sec^2 x - 1 \\
&= \int \tan^2 x \sec^2 x\,dx - \int \sec^2 x\,dx + \int dx && (\text{Let } u = \tan x. \\
&= \int u^2\,du - \tan x + x + C && \text{Then } du = \sec^2 x\,dx.) \\
&= \frac{u^3}{3} - \tan x + x + C = \frac{\tan^3 x}{3} - \tan x + x + C
\end{aligned}$$

(2) Let $u = \sec x$ and $dv = \sec^2 x\,dx$. Then $u = \sec x \tan x$ and $v = \tan x$.

$$\begin{aligned}
\int \sec^3 x\,dx &= \int \sec x \sec^2 x\,dx = \int u\,dv = uv - \int v\,du \\
&= \sec x \tan x - \int (\tan x)(\sec x \tan x)\,dx && \text{Integrate by parts.} \\
&= \sec x \tan x - \int (\sec^2 x - 1)\sec x\,dx && \tan^2 x = \sec^2 x - 1 \\
&= \sec x \tan x - \int \sec^3 x\,dx + \int \sec x\,dx
\end{aligned}$$

Combining the two secant-cubed integrals, gives

$$2\int \sec^3 x\,dx = \sec x \tan x + \int \sec x\,dx = \sec x \tan x + \ln|\sec x + \tan x| + C_1.$$

Therefore, $\int \sec^3 x\,dx = \frac{1}{2}(\sec x \tan x + \ln|\sec x + \tan x|) + C$.

(3)
$$\begin{aligned}
\int (\tan^4 x)(\sec^4 x)\,dx &= \int (\tan^4 x)(1 + \tan^2 x)(\sec^2 x)\,dx && \sec^2 x = 1 + \tan^2 x \\
&= \int (\tan^4 x + \tan^6 x)(\sec^2 x)\,dx && \text{Distribute.} \\
&= \int (u^4 + u^6)\,du = \frac{u^5}{5} + \frac{u^7}{7} + C && u = \tan x, \\
& && du = \sec^2 x\,dx \\
&= \frac{\tan^5 x}{5} + \frac{\tan^7 x}{7} + C
\end{aligned}$$

Products of Sines and Cosines

Integrals of the forms $\int \sin mx \sin nx \, dx$, $\int \sin mx \cos nx \, dx$ and $\int \cos mx \cos nx \, dx$.

Use the identities $\sin mx \sin nx = \frac{1}{2}\left[\cos(m-n)x - \cos(m+n)x\right]$

$$\sin mx \cos nx = \frac{1}{2}\left[\sin(m-n)x + \sin(m+n)x\right]$$

$$\cos mx \cos nx = \frac{1}{2}\left[\cos(m-n)x + \cos(m+n)x\right]$$

By combining some of the following identities, we obtain the above identities.

$$\sin(mx+nx) = \sin mx \cos nx + \cos mx \sin nx$$
$$\sin(mx-nx) = \sin mx \cos nx - \cos mx \sin nx$$
$$\cos(mx+nx) = \cos mx \cos nx - \sin mx \sin nx$$
$$\cos(mx-nx) = \cos mx \cos nx + \sin mx \sin nx$$

Example 8:
Evaluate $\int \sin 3x \cos 5x \, dx$.

Solution:

$$\int \sin 3x \cos 5x \, dx = \frac{1}{2}\int [\sin(3x-5x) + \sin(3x+5x)] \, dx$$
$$= \frac{1}{2}\int [-\sin(2x) + \sin 8x] \, dx$$
$$= \frac{1}{2}\left(\frac{\cos 8x}{8} - \frac{\sin 2x}{2}\right) + C$$

Remarks:

(1) $\int \sin mx dx = -\frac{\cos mx}{m} + C$

(2) $\int \cos mx dx = \frac{\sin mx}{m} + C$

(3) $\int \sin^2 x dx = \int \frac{1-\cos 2x}{2} dx$

(4) $\int \cos^2 x dx = \int \frac{1+\cos 2x}{2} dx$

(5) $\int \tan^2 x dx = \int (\sec^2 x - 1) dx$

(6) $\int \cot^2 x dx = \int (\csc^2 x - 1) dx$

(7) $\int \csc^2 x dx = -\cot x + C$

(8) $\int \sec^2 x dx = \tan x + C$